



Indian Institute of Technology Guwahati
Department of Physics
PH102
Tutorial-2

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1. If $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$ and $\vec{T} = 2yz\hat{x} + xy^2\hat{y} + x^2yz\hat{z}$, determine (a) $(\vec{\nabla} \cdot \vec{r})\vec{T}$, (b) $(\vec{r} \cdot \vec{\nabla})\vec{T}$
2. Evaluate the integral $\int_C r^2 d\vec{l}$, where, $r^2 = x^2 + y^2$, from the origin to the point P(1, 3) along the straight line connecting the origin to P(1, 3). $d\vec{l}$ is the differential vector in Cartesian coordinates.
3. Show that $\int_V \vec{B} \cdot (\vec{\nabla} \times \vec{A}) d\tau = \int_V \vec{A}(\vec{\nabla} \cdot \vec{B}) d\tau + \oint_S (\vec{A} \times \vec{B}) \cdot d\vec{a}$
4. Calculate the volume integral of the function $T = z^2$ over the tetrahedron with corners at (0, 0, 0) (1, 0, 0) (0, 1, 0) (0, 0, 1).
5. Check the fundamental theorem for gradients for the scalar field $T = x^2 + 4xy + 2yz^3$ the points $\vec{a} = (0, 0, 0)$, $\vec{b} = (1, 1, 1)$, and the three paths as shown in the figure 1 below.

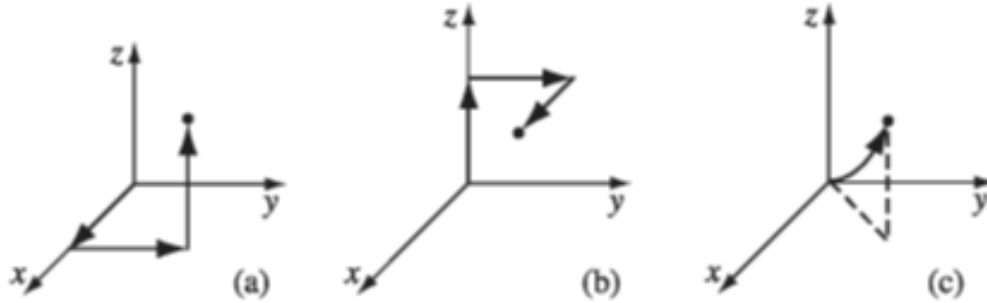


Figure 1

- (a) $(0, 0, 0)(1, 0, 0)(1, 1, 0)(1, 1, 1)$,
 - (b) $(0, 0, 0)(0, 0, 1)(0, 1, 1)(1, 1, 1)$,
 - (c) the parabolic path $z = x^2, y = x$
6. Check Stokes theorem using the formula $\vec{V} = ay\hat{x} + bx\hat{y}$ (a and b are constants) and the circular path of radius R centered at the origin in the xy plane.